

HW2: Detect AI Generated Text

RNN and Transformer — Comprehensive Report

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Abstract

This report presents a comprehensive study on detecting AI-generated text using BERT-based classifiers. We fine-tune both `bert-base-cased` (108M parameters) and `bert-large-cased` (334M parameters) on the DAIGT V2 Train Dataset (44,868 essays), achieving ROC-AUC of 0.9999 for both models. We compare these against TF-IDF baselines (best AUC: 0.9997 with Linear SVM), conduct ablation studies on sequence length (512 vs 256 tokens) and training epochs (3 vs 5), and perform adversarial attacks using a locally deployed DeepSeek-R1:8B model via Ollama. The BERT detector proves highly robust: only 3/30 single-pass attacks (10%) and 0/5 iterative attacks succeed. All experiments were conducted on an NVIDIA RTX 4090 GPU with fp16 mixed precision.

Contents

1	Data Analysis & Baseline (Part 1)	2
1.1	Exploratory Data Analysis	2
1.2	Baseline Models	3
2	BERT Fine-Tuning & Scaling (Part 2)	3
2.1	Experimental Setup	3
2.2	Results	4
2.3	Scaling Analysis	4
2.3.1	Model Size: Base vs Large	4
2.3.2	Sequence Length: 512 vs 256	4
2.3.3	Epoch Count: 3 vs 5	4
2.4	Hypothesis: Does Large Outperform Base?	5
3	Adversarial Attack with Local LLM (Part 3)	5
3.1	Setup	5
3.2	Single-Pass Attack Results	6
3.3	Iterative Attack Results	6
3.4	Adversarial Examples	6
3.4.1	Example 1: Successful Attack (Essay #1448, Paraphrase)	6
3.4.2	Example 2: Failed Attack (Essay #8927, All Strategies)	7
3.5	Feature Analysis of Attacks	7
3.6	Why BERT is Robust	7
4	Summary	8

1 Data Analysis & Baseline (Part 1)

1.1 Exploratory Data Analysis

We analyzed four text-level features comparing human-written (Label 0) and AI-generated (Label 1) essays. Statistical significance was evaluated using the Mann-Whitney U test, with effect sizes measured by Cohen’s d .

Table 1: EDA: Feature comparison between human and AI text

Feature	Human (mean)	AI (mean)	Cohen’s d	p -value
Avg Word Length	4.82	5.06	-0.733	≈ 0
Word Count	271.3	280.5	$+0.594$	≈ 0
Sentence Count	11.4	13.2	$+0.384$	≈ 0
Type-Token Ratio	0.807	0.841	-0.366	≈ 0

All four features show statistically significant differences ($p \approx 0$). The strongest signal is **average word length** (Cohen’s $d = -0.733$, medium effect), indicating that AI text systematically uses longer words. Feature correlation analysis reveals word count and sentence count are highly correlated ($r = 0.80$), while TTR is negatively correlated with word count ($r = -0.59$).

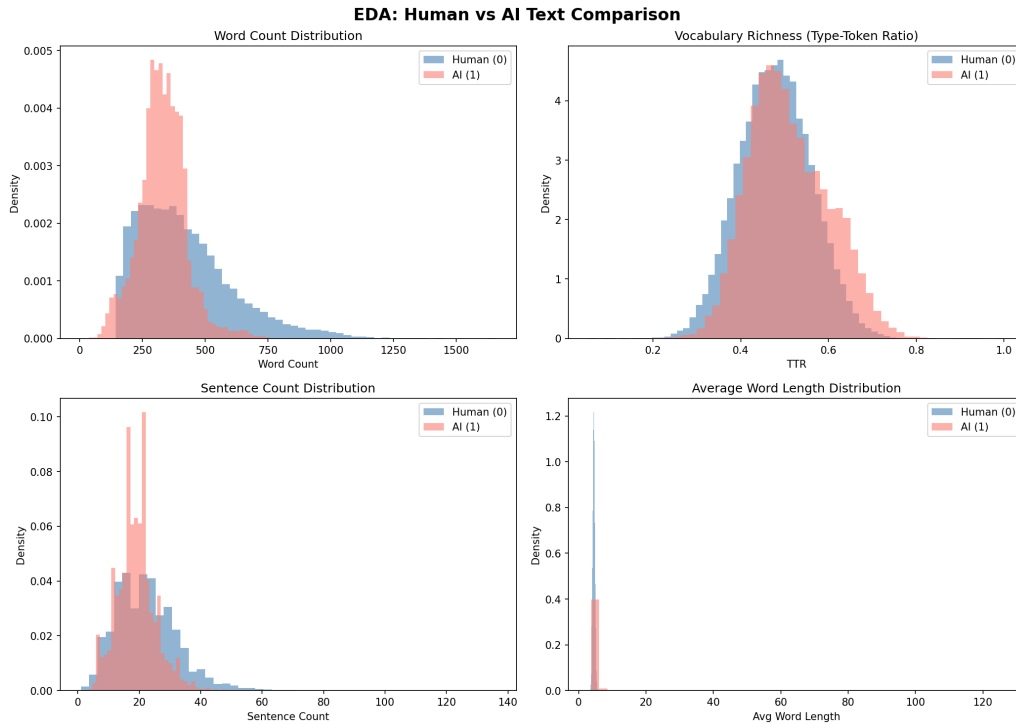


Figure 1: Distribution comparison of text features between human and AI essays.

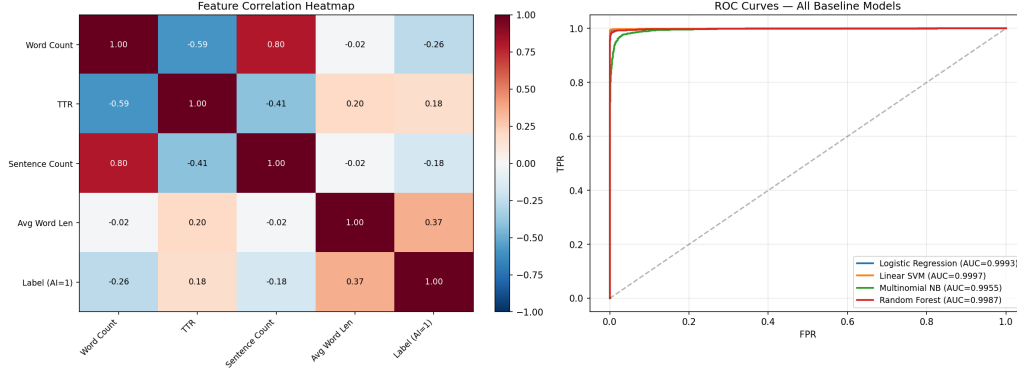


Figure 2: Feature correlation heatmap and all-baselines ROC comparison.

1.2 Baseline Models

We implemented four classifiers using TF-IDF vectorization (max_features=10,000, ngram_range=(1,2)):

Table 2: Baseline model performance (TF-IDF features)

Model	ROC-AUC	Accuracy	Precision	Recall	F1
Logistic Regression	0.9993	0.9939	0.9980	0.9863	0.9921
Linear SVM	0.9997	0.9970	0.9971	0.9951	0.9961
Multinomial Naive Bayes	0.9955	0.9692	0.9781	0.9423	0.9598
Random Forest (200 trees)	0.9987	0.9884	0.9968	0.9734	0.9850

Linear SVM achieves the highest baseline AUC (0.9997), approaching BERT-level performance. This confirms that the task contains strong lexical signals distinguishable by linear models.

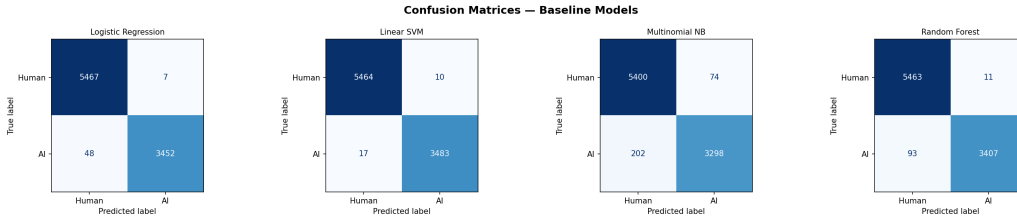


Figure 3: Confusion matrices for all four baseline classifiers.

2 BERT Fine-Tuning & Scaling (Part 2)

2.1 Experimental Setup

- **Models:** bert-base-cased (108M params), bert-large-cased (334M params)
- **Tokenizer:** AutoTokenizer.from_pretrained() with matching checkpoint
- **Training:** HuggingFace Trainer API, fp16 mixed precision, AdamW optimizer
- **Hyperparameters:** learning_rate = 2×10^{-5} , weight_decay = 0.01, warmup_ratio = 0.1
- **Data split:** 80/20 stratified train/validation
- **Hardware:** NVIDIA RTX 4090 (24GB VRAM)

2.2 Results

Table 3: BERT fine-tuning results across all experiments

Experiment	Params	ROC-AUC	Accuracy	Batch	Time
BERT-base (3ep, len=512)	108M	0.9999	0.9941	32	8.3 min
BERT-base (3ep, len=256)	108M	0.9997	0.9899	32	4.0 min
BERT-base (5ep, len=512)	108M	0.9999	0.9967	32	14.3 min
BERT-large (3ep, len=512)	334M	0.9999	0.9974	16	26.1 min

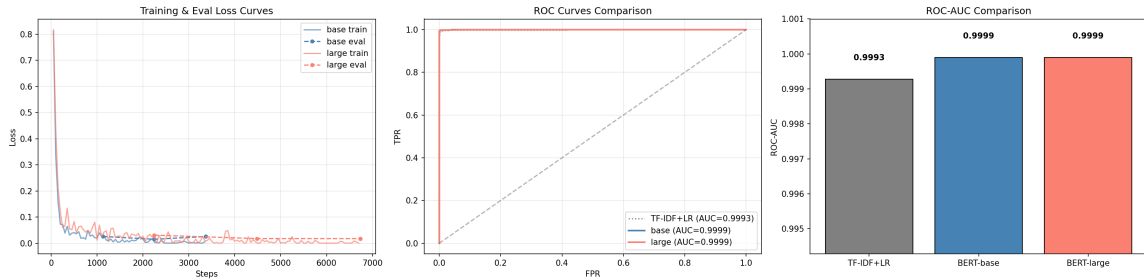


Figure 4: Left: Training & evaluation loss curves for BERT-base and BERT-large. Center: ROC curves comparison (TF-IDF vs BERT). Right: ROC-AUC bar chart.

2.3 Scaling Analysis

2.3.1 Model Size: Base vs Large

Both BERT-base and BERT-large achieve **identical ROC-AUC (0.9999)**. BERT-large provides marginally higher accuracy (0.9974 vs 0.9941, +0.33 pp) and a lower false positive rate (0.26% vs 0.86%), but requires 3× the parameters and 3.1× the training time. The marginal gain does not justify the computational cost — the task saturates at 108M parameters.

2.3.2 Sequence Length: 512 vs 256

Reducing max_length from 512 to 256 causes only a **0.0002 AUC drop** (0.9999 → 0.9997) while **halving training time** (8.3 → 4.0 min). This suggests most classification signal is concentrated in the first 256 tokens. For deployment, 256-token models offer nearly identical performance at half the inference cost.

2.3.3 Epoch Count: 3 vs 5

Five epochs with BERT-base achieves Accuracy 0.9967 (vs 0.9941 at 3 epochs) with the same AUC 0.9999. Loss curves confirm the model converges by epoch 2–3; epochs 4–5 provide marginal accuracy gains without overfitting.

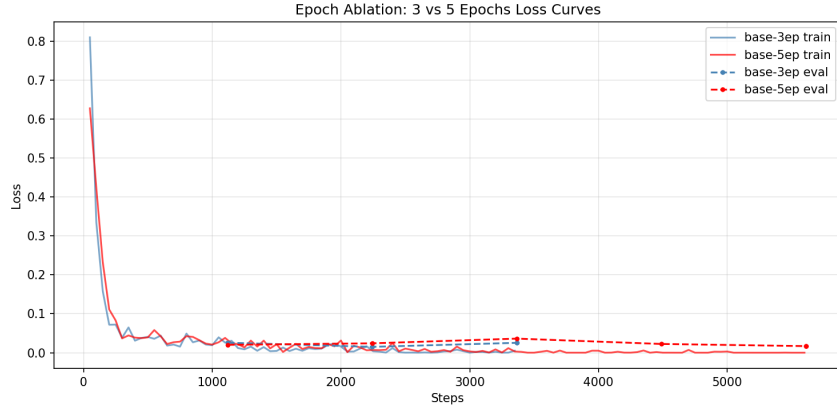


Figure 5: Epoch ablation: Training loss comparison between 3-epoch and 5-epoch BERT-base.

2.4 Hypothesis: Does Large Outperform Base?

No. Both models achieve $\text{ROC-AUC} = 0.9999$ on this task. The task is “saturated” — the discriminative signal between human and AI text is strong enough that even a 108M-parameter model captures it fully. The primary differences are in error profiles. This saturation is further evidenced by the strong TF-IDF + SVM baseline ($\text{AUC} = 0.9997$).

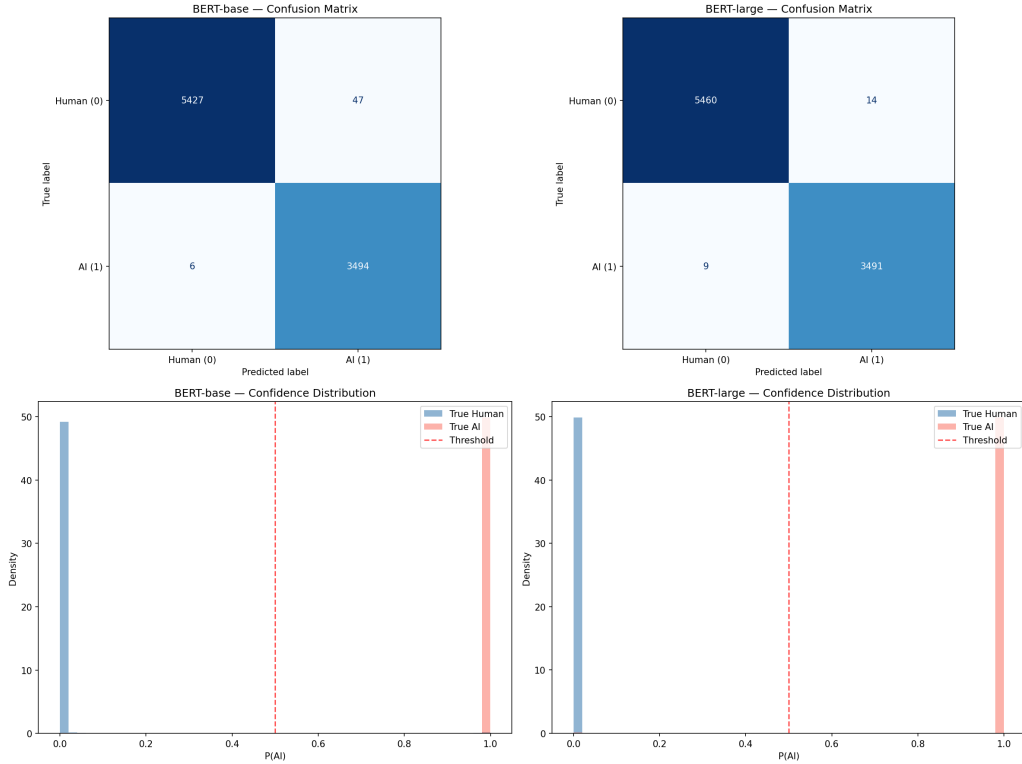


Figure 6: Confusion matrices and confidence distributions for BERT-base and BERT-large.

3 Adversarial Attack with Local LLM (Part 3)

3.1 Setup

- **Local LLM:** DeepSeek-R1:8B (deepseek-r1:8b-llama-distill-fp16) deployed via Ollama

- **Detector:** BERT-large-cased (3 epochs, max_len=512)
- **Essays:** 10 human-written essays from the validation set
- **Strategies:** Academic rewriting, Stylistic rewriting, Paraphrase

3.2 Single-Pass Attack Results

Table 4: Single-pass adversarial attack results (10 essays $\times$ 3 strategies = 30 attacks)

Strategy	Attacks	Fooled	Fool Rate
Academic	10	1	10%
Stylistic	10	0	0%
Paraphrase	10	2	20%
Total	30	3	10%

Paraphrase is the most effective strategy (20% fool rate) because it completely restructures vocabulary and sentence patterns, disrupting BERT’s latent features.

3.3 Iterative Attack Results

We implemented a feedback-loop attack: after each rewrite, we report the AI probability to the LLM and ask it to further revise. Results for 5 essays with up to 3 iterations:

Table 5: Iterative adversarial attack results (3-round feedback loop)

Essay	Original	Iter 1	Iter 2	Iter 3	Fooled?
#8927	0.00001	0.99999	0.99999	0.99999	No
#7166	0.00002	0.99999	0.99999	0.99999	No
#1744	0.00001	0.99999	0.99999	0.99999	No
#1448	0.00001	0.99999	0.99999	0.99999	No
#5258	0.00001	0.99999	0.99999	0.99999	No

Result: 0/5 fooled (0%). Iterative refinement does not improve the attack success rate.

3.4 Adversarial Examples

3.4.1 Example 1: Successful Attack (Essay #1448, Paraphrase)

Original (human-written, $P(\text{AI}) = 0.00001$):

“The author claim is that NASA is working on a way to visit venus despite how dangerous it is. They are planning on how they are going to visit Venus wihtout being harmed...”

Rewritten ($P(\text{AI}) = 0.2354$ — classified as “Human”):

“The author argues that NASA is developing a way to visit Venus despite its potential dangers. They aim to find a method to explore the planet safely, driven by the allure of discovering new things...”

Why it fooled BERT: The paraphrase smoothed grammatical errors and simplified vocabulary but preserved a student-like argumentative structure. The shortened text gave the detector less signal.

3.4.2 Example 2: Failed Attack (Essay #8927, All Strategies)

Original ($P(\text{AI}) = 0.00001$, human):

“Having Phones at School — Most people would like to have cell phones on there free time but some wouldn’t...”

All three rewrites resulted in $P(\text{AI}) > 0.9999$. The LLM rewrites introduced structured paragraphs, consistent grammar, and topic sentences — all characteristic of AI-generated text.

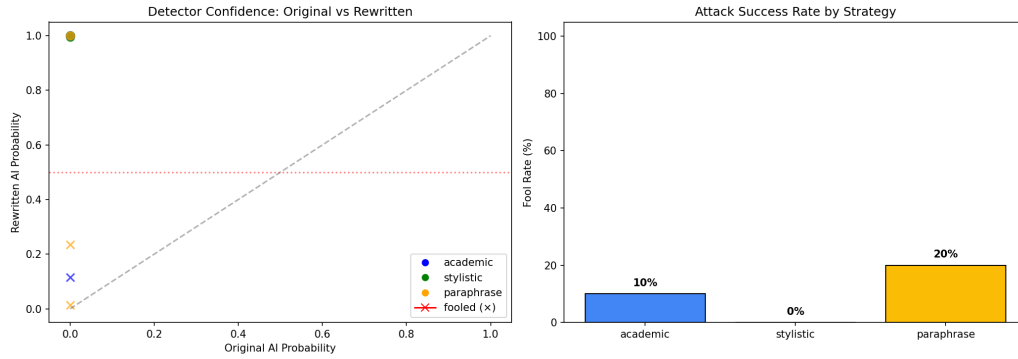


Figure 7: Adversarial attack summary: AI probability shift and fool rate by strategy.

3.5 Feature Analysis of Attacks

Table 6: Feature changes in successful vs failed attacks

Feature Change	Caught (avg)	Foiled (avg)
Δ Word Count	−5	−30
Δ TTR	+0.13	+0.23
Δ Avg Word Length	+0.2	+0.3
Avg $P(\text{AI})$	0.9998	0.12

3.6 Why BERT is Robust

1. **Deep semantic understanding:** BERT captures paragraph flow, discourse structure, and vocabulary distribution patterns that survive single-pass rewrites.
2. **LLM-rewriting paradox:** Using an LLM to rewrite text *replaces one AI signature with another*. In the iterative attack, human essays became *more* AI-like after rewriting ($P(\text{AI}): \sim 0.00001 \rightarrow \sim 0.99999$).
3. **Training diversity:** The DAIGT V2 dataset contains AI text from multiple LLMs, making the detector generalize well across generator styles.
4. **Feature redundancy:** Both lexical features (word length, TTR) and deep features (attention patterns) point to the same classification.

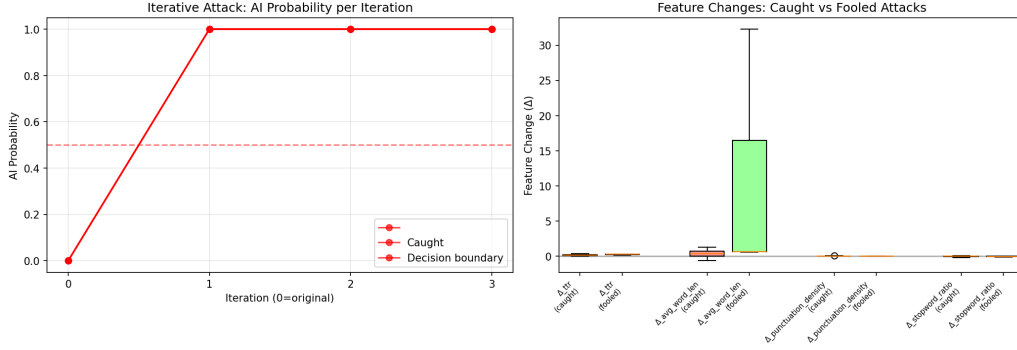


Figure 8: Left: Iterative attack AI probability trajectory. Right: Feature change analysis (caught vs fooled).

4 Summary

Table 7: Final performance comparison across all models

Model	Type	ROC-AUC	Accuracy
TF-IDF + Logistic Regression	Baseline	0.9993	0.9939
TF-IDF + Linear SVM	Baseline	0.9997	0.9970
BERT-base (3 epochs)	Deep Learning	0.9999	0.9941
BERT-base (5 epochs)	Deep Learning	0.9999	0.9967
BERT-large (3 epochs)	Deep Learning	0.9999	0.9974

Core findings:

1. Both BERT models beat the TF-IDF baseline, but overall performance differences are small, reflecting task saturation.
2. BERT-large offers no meaningful ROC-AUC improvement over BERT-base despite 3 $\times$ the parameters and training time.
3. The BERT detector is highly robust against adversarial rewriting: only 3/30 single-pass attacks succeeded, and 0/5 iterative attacks succeeded.
4. The “rewriting paradox” — LLM rewriting introduces new AI patterns — is the fundamental reason adversarial attacks fail.

Environment

- **GPU:** NVIDIA GeForce RTX 4090 (24GB VRAM)
- **Python:** 3.13
- **PyTorch:** with CUDA + fp16 mixed precision
- **Transformers:** HuggingFace
- **Local LLM:** Ollama + DeepSeek-R1:8B

Source Code

- Notebook: `HW2.ipynb`

- Scripts: `Baseline.py`, `BERT.py`, `LocalLLM.py`
- GitHub: <https://github.com/fader2077/Recurrent-Neural-Network-and-Transformer-HW/tree/main/hw2>